

technical opening

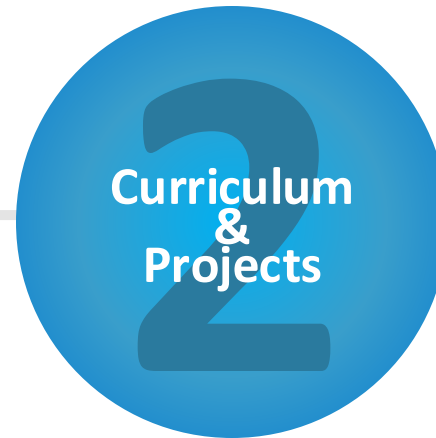
presentation

ICT | Cyber Security Master Program

Tech Meeting ...



Agenda for ISM



Bucharest, Romania, E.U.





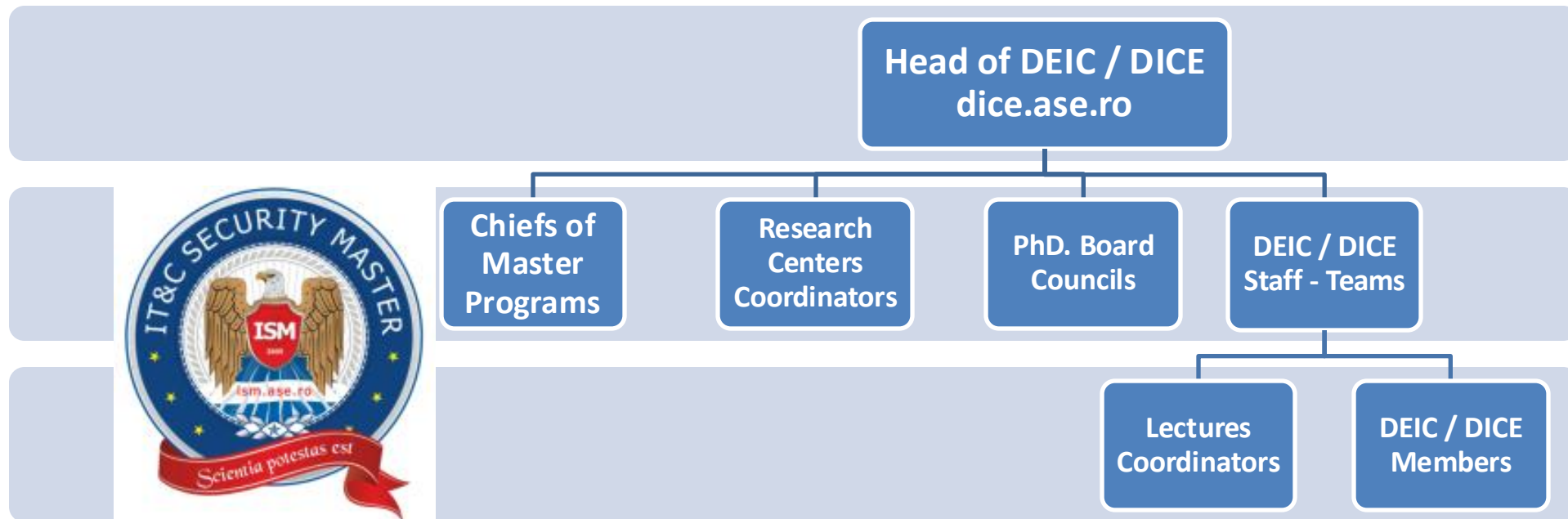
Organization Form, Statistics, Students Structure , Target Group Profiles

ISM Features & Organization Form



1.1 Organization Form

Main **components** to kick-off our discussion –
ISM Master Program, Department, Faculty, University
ISM | ACS Team – Board of Directors
www.acs.ase.ro



<http://ism.ase.ro>

1.1 Organization Form

ISM – ICT | Cyber Security Master | Informatics Security Master

- **Website:** www.ism.ase.ro
- **Duration:** 4 semesters
- **Motto:** Knowledge is power
- **Lectures/Seminars/Labs** are structured in modules during – the weekend Exams Sessions in the end of each semester + one re-exams session /year
- **THE DIPLOMA** issued by *The Bucharest University of Economic Studies* for **Informatics SECURITY MASTER – in English**, grant access to PhD. programs from EU & US



1.1 Organization Form

ISM – IT&C Security Master / Informatics Security Master

- **START in 2005 & it is accredited by ARACIS** since 2008
- **Since 2013,**
 - **according to the Education Law from 2011**, published in the Official Journal of Romania no. 18/10.01.2011,
 - **the master programs is accredited by ARACIS according to the ARACIS Council Decision from 26-27.06.2013 (renewed in 2018) | http://www.aracis.ro/uploads/media/Hotarari_ale_Consiliului_26-27.06.2013.pdf)** in terms of:
 - Name: **IT | ICT | Cybersecurity Master | Securitatea Informatica (cu predare in limba engleza)**
 - Curriculum: **www.ism.ase.ro**
 - Location: **Bucharest**
 - Language: **English**
 - Students Number: **45 + 5 (max 70)**



1.1 Organization Form

ISM – IT | ICT | Informatics | Cyber Security Master Program **is 100% Online & Digital**

- * Video conferences platforms for lectures and labs: **Zoom, Teams, Meet**
- * Communications: www.ism.ase.ro, **Slack** / **Teams** / **Discord**, **Google Calendar**
- * Social Networks: **Twitter/X**, **LinkedIn (Alumni?)**, **Facebook**, **Instagram**, ...
- * Code and Storage: **GitHub** & **Google Drive** & **YouTube**
- * Learning Management Systems: www.online.ase.ro -> <https://student.ism.ase.ro/portal/>



- **ISM** provides the environment for PhD. stages in E.U & U.S + Journals Publishing + Business Development
- **Moodle/SAKAI** – PHP/Java EE open source based projects were the ISM LMS
- **ISM's** main target of the '**I18N - Internationalization**' is to have competent foreign students from world wide + students & professors mobility
- **Cyber-Security Webinar/World-Class in English**

1.1 Organization Form - CONNECTIVITY

ISM – IT | ICT | Informatics | Cyber Security Master Program ... 100% Online & Digital? vs. presence campus!

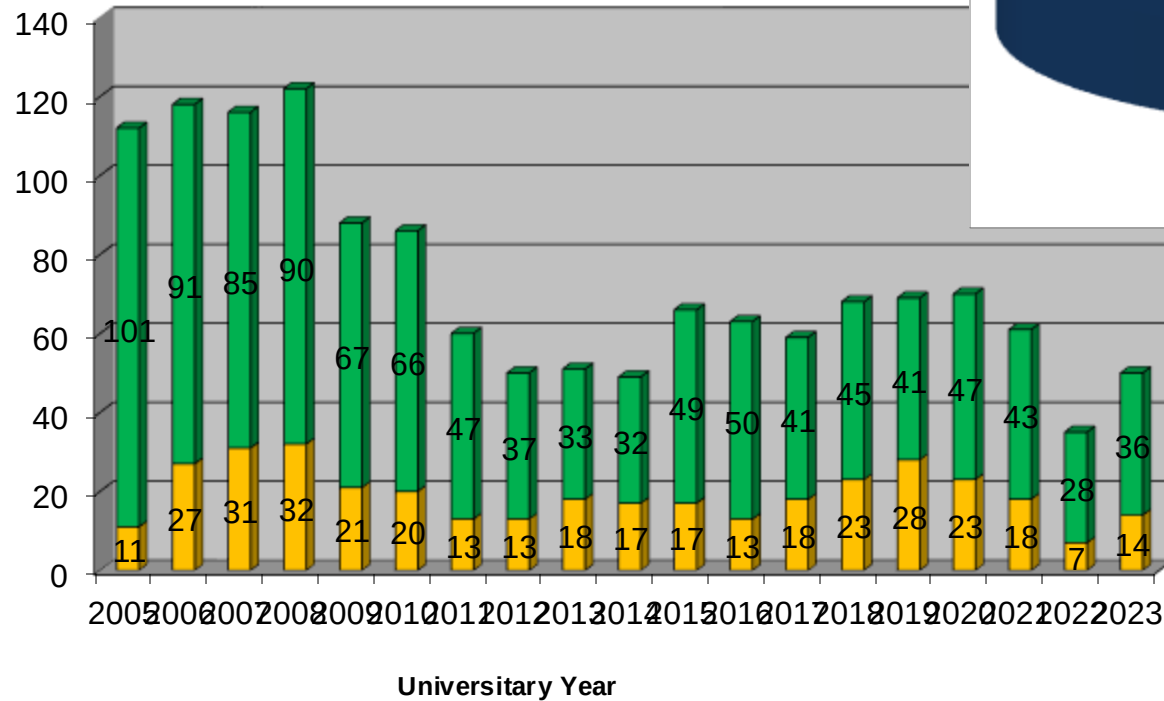
- * **Website:** <http://ism.ase.ro>
- * **E-mail:** ism.ase.ro@gmail.com | ism@ase.ro
- * **Slack** messaging/communication platform: <https://ism-it-sec-2024-2026.slack.com/>
- * **Google calendar:**
<https://calendar.google.com/calendar/embed?src=ead748160275e99a5c445210d18a5a0200a6c6963fd62a8e042c445fb440584a%40group.calendar.google.com&ctz=Europe%2FBucharest>
- * **Zoom video-conference:** the connection link (for the ones who are not able to physically attend) will be sent before the lectures and labs activities via Slack and in the worst case scenario via e-mail.
- * **LMS:** Moodle – <https://portal.ism.ase.ro> <- Sakai - <https://student.ism.ase.ro/portal/>
- * **International Conference website:** www.secitc.eu



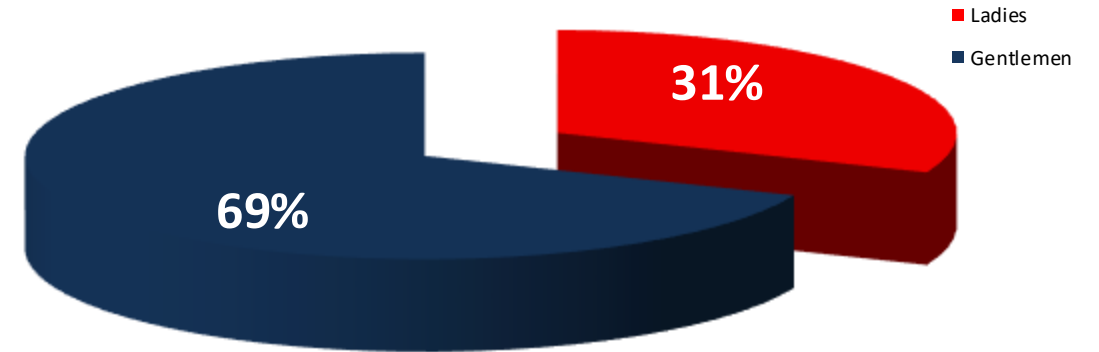
1.2 Statistics – Students Structure

ISM – ICT | Cyber Security Master

Students Number Structure Statistics - 2005-2008 eligible students
2009-2023 enrolled students



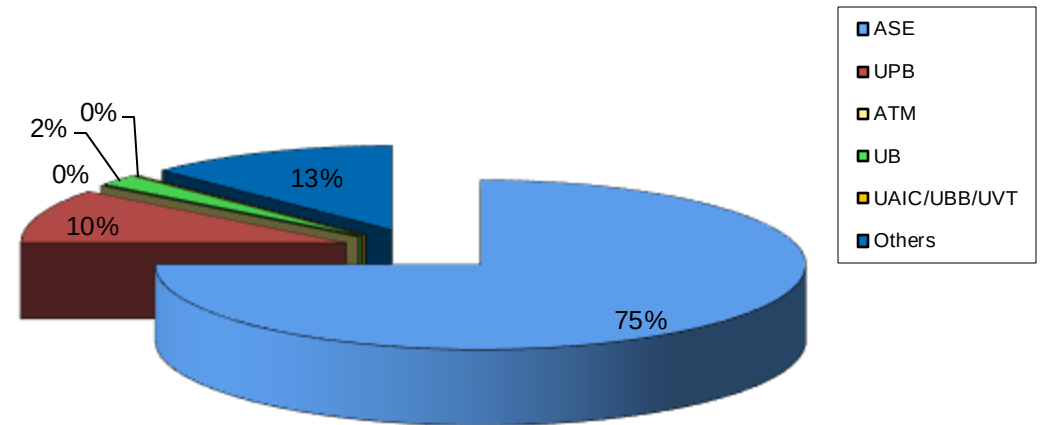
Students 2024
48 students



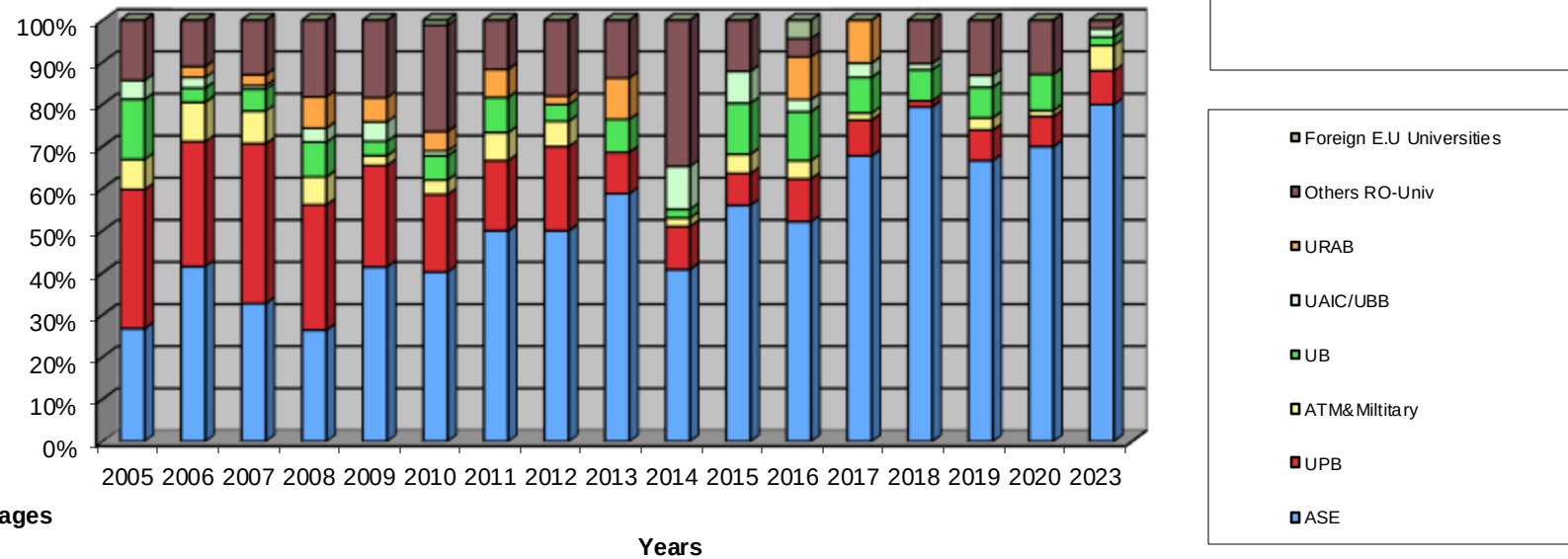
1.2 Statistics – Students Structure

ISM – ICT | Cyber Security Master

Students Structure by Universities in 2024



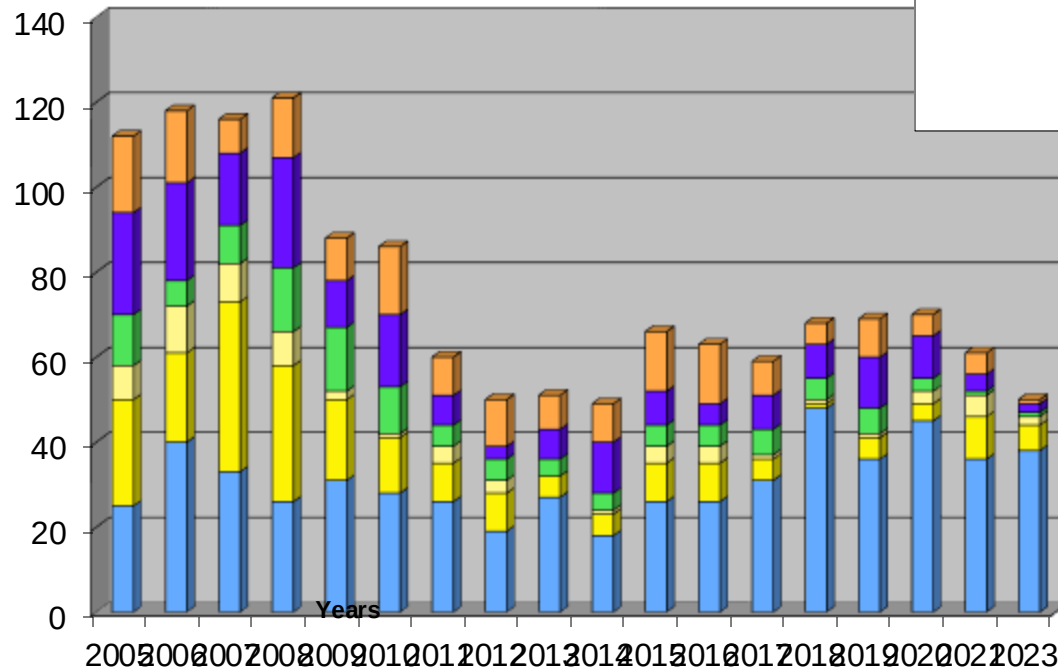
Universities Statistics - 2005-2008 eligible students/2009-2023 registered students



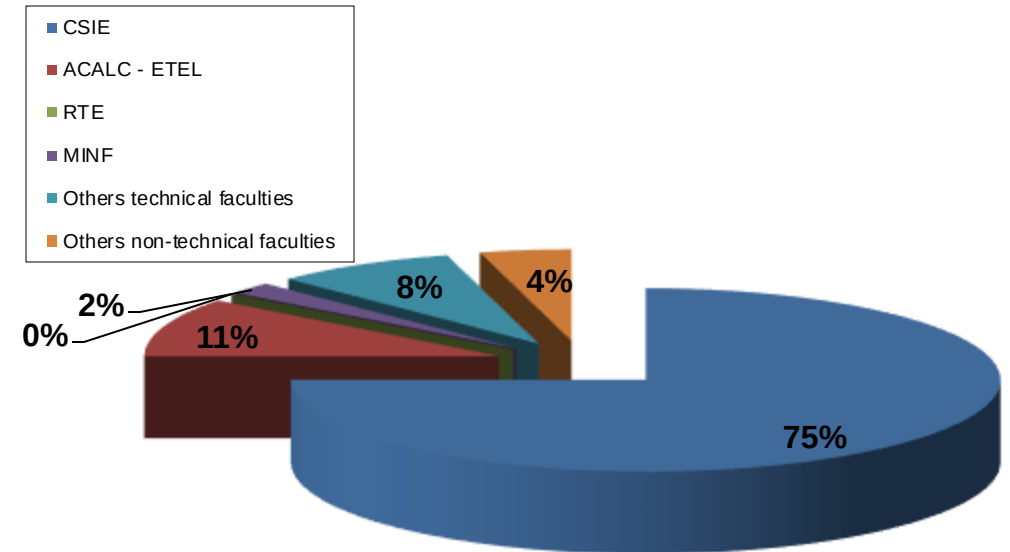
1.2 Statistics – Students Structure

ISM – ICT | Cyber Security Master

Students BSc. Programs - 2005-2008 eligible students/2009-2023 registered students



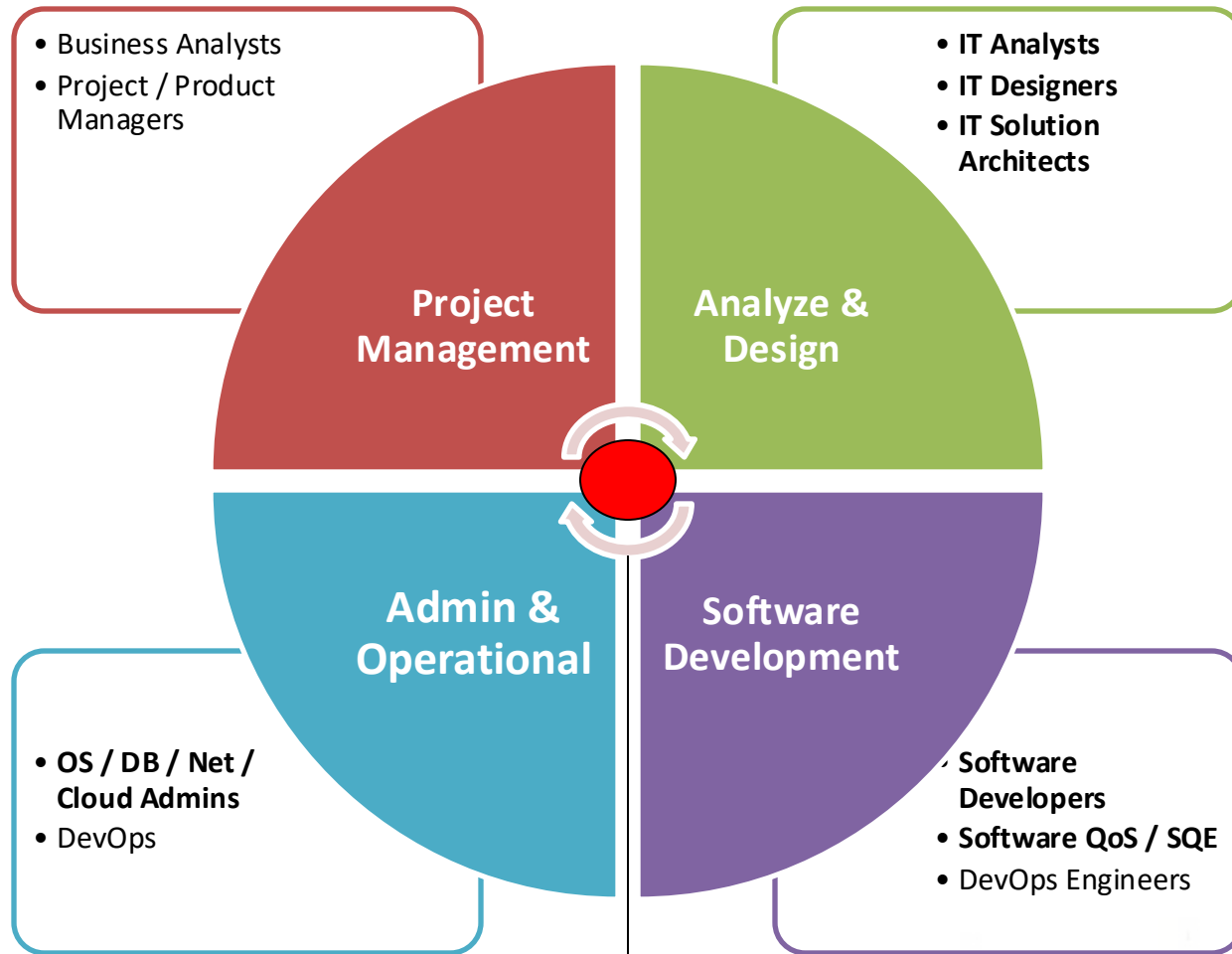
Students 2024 BSc. Programs



- Others non-technical faculties
- Others technical faculties
- MINF
- RTE
- ACALC - ETEL
- CSIE

1.3 Target Group Profile

ISM – Cyber Security | ICT | Informatics Security Master



Security Analyst
Security Engineer
Security Architect
Security Administrator
Security Software Developer
Cryptographer
Cryptanalyst
Security Consultant

ICT Security Solution Architect
CISO – Chief Information Security Officer

Is what you expected?

This was **Section 1 – ISM Technical Issues**

- **Multiple IT/Informatics/Information/Cyber Security Master Programs** in the RO/EU/EMEA/US and JPAC
- Mark **8/10 @ UPB is NOT** the same with **8/10 @ Faculty of Law** => The **NECESSITY** to keep the **ADMISSION EXAM** from C/C++ & Java (and basic networking?) for everybody (we saw medics-doctors good IT Dev/Pro BUT not always) – how will be the exam for the foreign students ?
- The **optimal number for ISM is 60** students with 90% IT specialists.
- **ISM is READY to be 'mobile' / world-wide accessible**, but SHOULD be 'PRIVATE-PUBLIC' a.s.a.p with Google/Oracle/Microsoft/IBM support (2011) and MUST provide scholarships?, in order to have students from aboard – **ISM – IT | Cyber Security Master Program is enabled for 100% digital and online ...**
- **ISM already have Oracle, ISACA, DELL, ... support – Guest Speakers for 20% didactical activities (since 2013)**

Section Conclusion

ISM Motto: **Knowledge is Power**

ISM is an IT&C technical master program since 2005, it will be in English with I18N mission, technological transfer – to the students + to/from companies; it's all about multi-disciplinary work, and IT **development & integration** tasks.





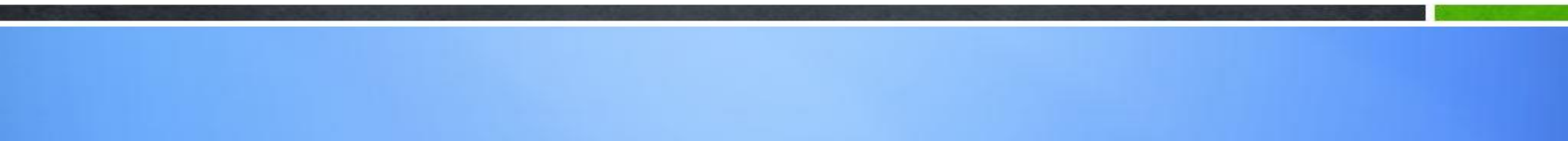
Projects: ISM, SECITC Conference, ISH - Informatics Security Handbook, JITCS/PITCS, JMEDS, SECITC Community

ISM Curriculum & ISM-SECITC Projects



It's not just about the ideas, but implementations

ISM Curriculum & Projects



What about the **Curriculum & Projects**
Technological Transfer & Issues?

Some of them are new.

2.1 ISM Curriculum – Dev Platforms on TIOBE

Very Long Term History

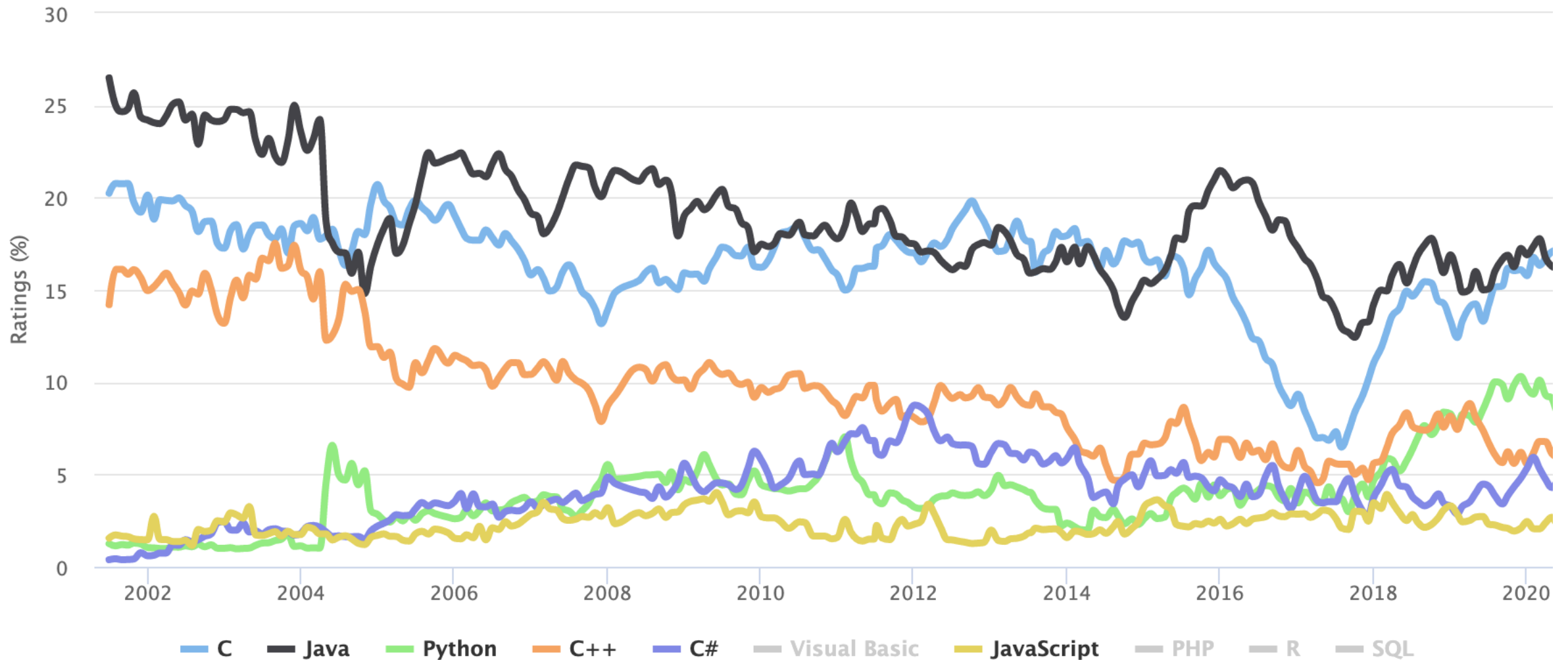
To see the bigger picture, please find below the positions of the top 10 programming languages of many years back. Please note that these are *average* positions for a period of 12 months.

Programming Language	2021	2016	2011	2006	2001	1996	1991	1986
C	1	2	2	2	1	1	1	1
Java	2	1	1	1	3	15	-	-
Python	3	5	6	8	25	24	-	-
C++	4	3	3	3	2	2	2	6
C#	5	4	5	7	13	-	-	-
Visual Basic	6	13	-	-	-	-	-	-
JavaScript	7	7	10	9	9	20	-	-
PHP	8	6	4	4	10	-	-	-
SQL	9	-	-	-	37	-	-	-
Assembly language	10	11	-	-	-	-	-	-
Ada	31	28	17	16	18	7	3	2
Lisp	33	27	13	13	17	8	6	3
(Visual) Basic	-	-	7	5	4	3	5	5

2.1 ISM Curriculum – Dev Platforms on TIOBE

TIOBE Programming Community Index

Source: www.tiobe.com



2.1 ISM Curriculum – Year 1, Semester 1

No.	Course Name	Type	Credits	Evaluation	Hours	Course	Laboratory
1	Cryptography Basis	O	5	E	28	14	14
2	Electronic Signature	O	5	E	28	14	14
3	Security Standards and Protocols	O	5	E	28	14	14
4	Antivirus and Virus Technologies	O	5	E	42	28	14
5	Secure Applications Programming	O	5	E	70	42	28
6	ICT Security Systems Audit, Laws and Regulations	O	5	E	28	0	28
	TOTAL of the Mandatory Lectures		30	6E	224	112	112
	OVERALL TOTAL		30	6E	224	112	112

2.1 ISM Curriculum – Year 1, Semester 2

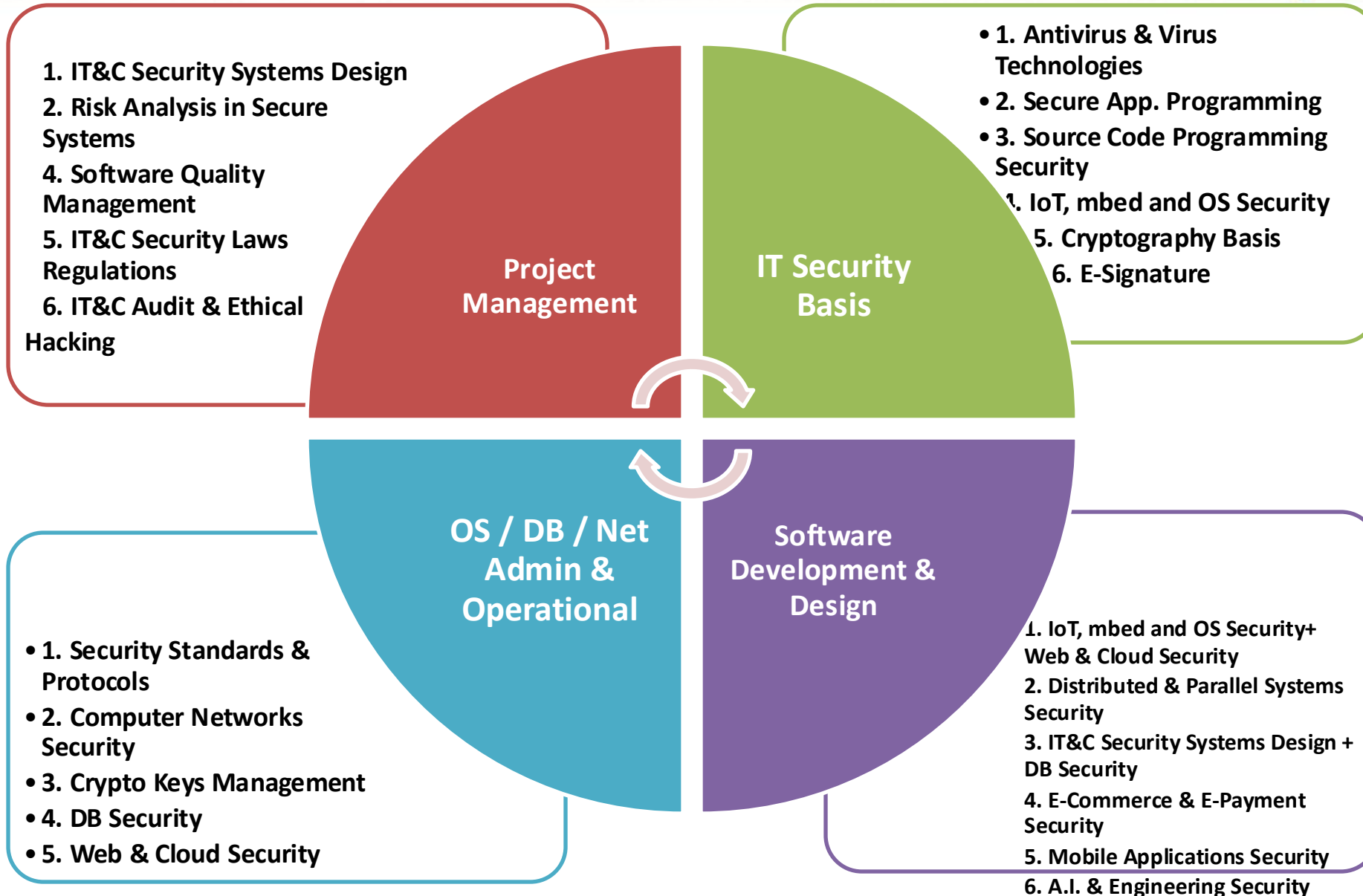
No.	Course Name	Type	Credits	Evaluation	Hours	Course	Laboratory
1	Computers Networks Security	O	4	E	28	14	14
2	Distributed and Parallel Systems Security	O	4	E	28	14	14
3	Databases Security	O	4	E	42	14	28
4	Web and Cloud Applications Security	O	6	E	56	28	28
5	Blockchain, Quantum Cryptography and E-Payment / E-Commerce Security	O	4	E	28	14	14
6	Multimedia Computing Security and Biometrics	O	4	E	28	14	14
7.1	Cryptographic Keys Management	A	4	E	14	14	0
7.2	Security Management in Informatics Systems	A	4	E	14	14	0
	TOTAL of the Optional Lectures		4	E	14	14	0
	TOTAL of the Mandatory Lectures		26	6E	210	98	112
	OVERALL TOTAL		30	7E	224	112	112

2.1 ISM Curriculum – Year 2, Semester 1

No.	Course Name	Type	Credits	Evaluation	Hours	Course	Laboratory
1	Source Code Programming Security	O	6	E	28	14	14
2	Robotics, IoT (Internet of Things) & Embedded Operating Systems Security	O	4	E	42	28	14
3	Mobile Applications Security	O	4	E	42	14	28
4	Privacy, Ethical Hacking and Digital Forensics	O	4	E	28	14	14
5	Artificial Intelligence in IT Security Systems	O	4	E	28	14	14
6	Hardware Security – FPGA and TEE – Trusted Execution Environments	O	4	E	28	14	14
7	Software Quality Management & Assurance	O	4	E	28	14	14
	TOTAL of the Mandatory Lectures		30	7E	224	112	112
	OVERALL TOTAL		30	7E	224	112	112

* Year 2, Semester 2: ICT Security Systems Engineering, Design and DevOps

2.1 ISM Curriculum – Disciplines per Target Groups



2.2 ISM-SECITC Projects

1. ISM Web Site + LMS: SAKAI/Moodle

An electronic platform for students.



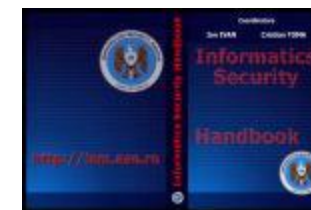
2. SECITC Conference

An IT&C Security conference for academic staff & companies.



3. ISM-SECITC Dissemination

The students' opportunities to publish their work and research.



4. ISM-SECITC Community

Electronic platforms for IT&C social networking.



2.2 ISM Website + Google Calendar



Google calendar

1. Visit Google Calendar:
Synch your Mobile
Phones, E-Mail
Agents, Laptops



2. Follow Social Networks
@ LinkedIn, Facebook &
X/Twitter @
twitter.com/ismasero :
Important tweets
REPLACED by
Slack/LinkedIn?

ism@ase.ro (+40) 021 319 19 00 / int 310 f t in Search here..

Home About Admission Curricula Schedule Student News Contacts

IT&C SECURITY | CYBERSECURITY MASTER
Department of Economic Informatics and Cybernetics
Scientia potestas est

IT&C Security
The master is recognized by students, academia and industry as one of the best MSc master programs on IT&C Security and Cybersecurity in

MSc Program
Since 2005 we have been educating professionals in the area of the information technology and communications security. In

Academic Partnership
The program academic staff gathers well known professors and specialist from partner universities and from the IT industry

2.2 SECITC Conference

SECITC – <http://www.secitc.eu>

see what we can do for you.

Cyber | IT&C Security Master
Bucharest University of Economic Studies

Information Technology Security Master
Military Technical Academy

Advanced Technologies
Institute



HOME COMMITTEES CALL FOR PAPERS PROGRAM K-SPEAKERS REGISTRATION INFO CONTACT



Springer



2.2 SECITC Conference

- **SECITC Conference:** International Conference on Security for Information Technology and Communications – YEARLY in November in Bucharest – *started from 2006 by ISM – IT&C Security Master* and *continued together* in partnership with Military Technical Academy – Information Security Master Program *from 2008*
- **Goal:** to bring together specialists that are researching & developing/designing solutions in the field of IT&C Security. The event offers to engineers, software developers, mathematicians and scientists, the possibilities to establish new collaborations within current and future research projects, to present research results, to establish academic collaborations, to link the university research with the industry.
- **Results:** proceedings publishing in JITCS – Journal of Information Technology and Communications Security, ISH – Informatics Security Handbook , JAQM – Journal of Applied Quantitative Methods – www.ism.ase.ro, www.secitc.eu, www.jaqm.ro.
- **Participants:**
 - Professors from IT&C Security field - **UNIVERSITIES.**
 - Specialists from IT&C companies – **COMPANIES.**
 - Students from master and PhD. programs – **STUDENTS.**

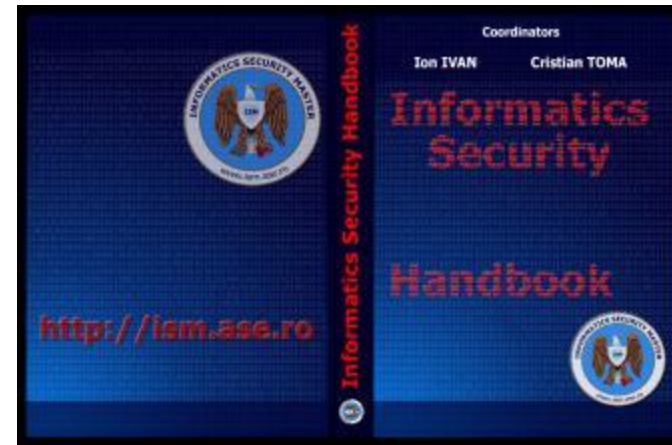
2.2 ISM-SECITC Dissemination

ISH: Informatics Security Handbook –
www.ism.ase.ro

JMEDS: Journal of Mobile, Embedded and
Distributed Systems – www.jmeds.eu

JITCS: Journal/Proceedings of Information
Technology and Communications Security
Conference – www.secitc.eu

JAQM: Journal of Applied Quantitative Methods –
www.jaqm.ro



2.2 ISM-SECITC Community

+

Find ISM, SECITC Conference & SECITC Community @

Twitter.com - X / LinkedIn.com / Facebook.com



an on-line Internet Community – Social Network

**Facebook helps you connect and share with
the people in your life.**





ISM Issues Summary
for easy sharing

Section Conclusions

2.1 ISM Curriculum

2.2 ISM & SECITC Projects

- ISM web
- SECITC Conference
- ISM-SECITC Dissemination
 - ISH
 - JMEDS
 - JITCS / PITCS
 - JAQM
- ISM-SECITC Community



Share knowledge, Empowering Minds

Communicate & Exchange Ideas



Questions & Answers!

But wait...

There's More!



ISM Roadmap – Goals vs. Achievements

» Details about ISM-Roadmap

- **Informatics Security Handbook 3rd Edition** – ISH for ISM
- **I18N** + students/professors mobility & interdisciplinary collaboration with various universities – **ISM & SECITC Conference – partially done** – ERASMUS?/how to provide scholarships for foreign students in order to collaborate with the for the research grants?
- **IT Companies involvement** – ISM & SECITC Conference – **DONE with Oracle & Microsoft**, discussions started with other companies – Academic Program @ University level - **AMCham**
- **Students publishing** papers in conferences proceedings and workshops – ISM, SECITC Conference => in **JITCS/PITCS, ISH, JMEDS & JAQM - DONE**
- **Students involvement in research and ISM activities** – in **ISM activities – DONE**, ... what research?

ISM Roadmap – Goals vs. Achievements

» Details about ISM: **Roadmap – it might be 100% Digital & Online**

- * Video conferences platforms for lectures and labs: **Zoom, Teams, Meet**
- * Communications: www.ism.ase.ro, **Slack / Teams, Google Calendar**
- * Social Networks: **Twitter, LinkedIn, Facebook, Instagram, Tik-Tok - LOL?, ...**
- * Code and Storage: **GitHub & Google Drive & YouTube**
- * Learning Management Systems: www.online.ase.ro -> <https://portal.ism.ase.ro>

- **IMPROVEMENT of Digital/Online Platforms** for: Professors audit, Online student platform for exams, bookmarks and assignments, *students job tracking* indicator – ISM web site & Moodle/SAKAI platform – partially done.
- **Improvement of lectures** – continuous process
- **Students selection** – MUST keep the ADMISSION EXAM for all budget and tax places & English interview over Internet (Zoom/Teams/Meet/Skype/...)
- **Students selection** – MUST have at least 2 or 3 disciplines per semester with PC tests beyond the quiz tests in Moodle/Sakai | Online evaluation with proctoring
- **Hardware & Software Acquisitions vs. CLOUD** – ISM needs **Private CLOUD & IoT / Robotics?**
- **Web site improved** including web form for **collecting companies “wish list”** for delegating projects to university using the external research resources
- **On-line shop with ISM products** – pens, mugs, t-shirts – like in Google Shop



Questions & Answers!

But wait...

There's More!

Students Poll

<https://forms.gle/g8Vk9rxbz7HYCG1h7>

ISM Conclusions

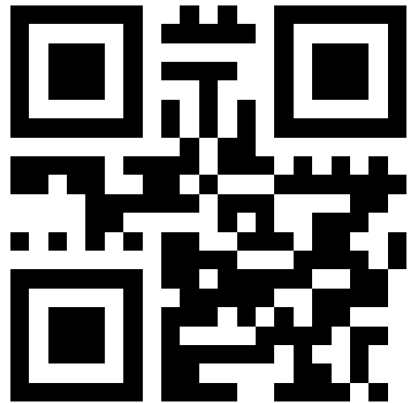
» Details about ISM - Conclusions

- **I18N & Open Master Program** – collaborations with ATM, UPB, UB, UBB Cluj, UVT
- **Most known didactic specialists invited** as associated professors – Victor Valeriu PATRICIU, Ion BICA, Mihai PURA (ATM), Mihai Chiroiu (UPB), ...
- **Interdisciplinary Master Program** with heterogeneous students
- **Company involvement** – guest speakers from Oracle, ISACA, banks, ... it will be Microsoft, IBM & Google again?
- **Transparent & permanent improvement**
- **Interactive & dynamic and it might be 100% digital & online**

www.ism.ase.ro

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Mulțumesc!